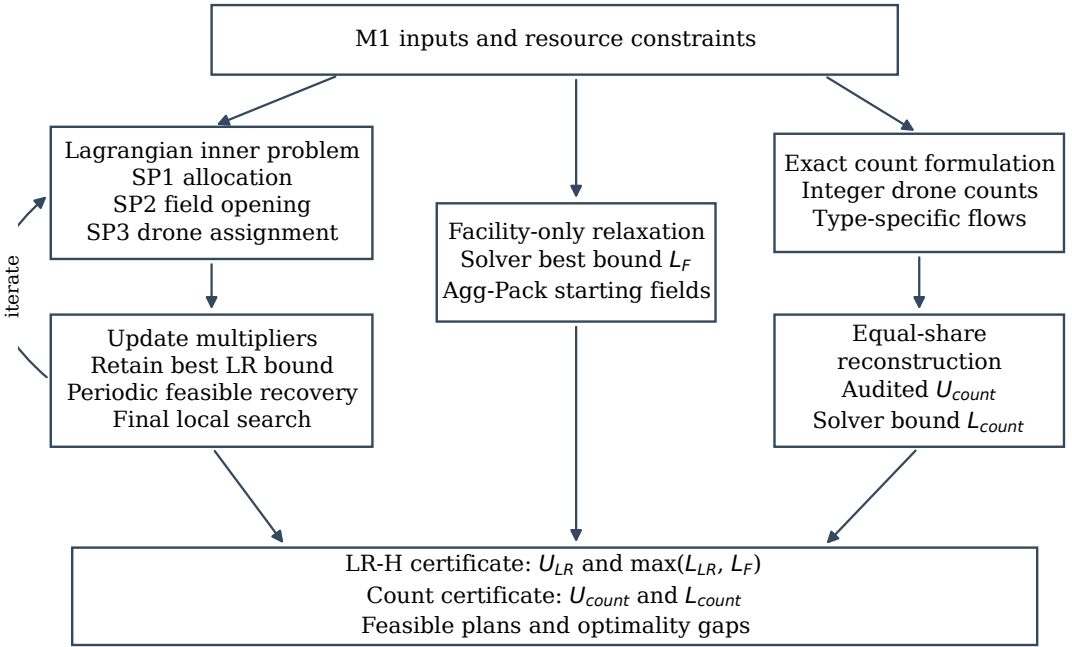
M1 inputs and resource constraints
Lagrangian inner problem
SP1 allocation
SP2 field opening
SP3 drone assignment
Facility-only relaxation
Solver best bound L_F
Agg-Pack starting fields
Exact count formulation
Integer drone counts
Type-specific flows
iterate
Update multipliers
Retain best LR bound
Periodic feasible recovery
Final local search
Equal-share reconstruction
Audited U_{count}
Solver bound L_{count}
LR-H certificate: U_{LR} and max(L_{LR}, L_F)
Count certificate: U_{count} and L_{count}
Feasible plans and optimality gaps